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Econ 310

Examining the Effect of TikTok on Song Duration

1. Introduction

There is no doubting the effect that the app *TikTok* has had on the way that media is produced and consumed since its launch in the U.S. in 2018. Short-form content has made its way into almost every social media app and has changed the way many people receive entertainment. There has been talk online that modern popular music artists are catering towards being easily formatted into short-form content, which has had the effect of shortening songs in general. I decided to test this out. I hypothesize that the most popular songs (Billboard Top 100) in the U.S. have gotten shorter since the launch of TikTok in 2018. The relevant population data would be any song that had charted within the year on the Billboard Hot 100 in the U.S., which is based on sales (physical and digital), online streaming, and radio airplay in the U.S. This gives us an idea of what the most popular songs for an entire year in the United States. I decided to choose the past 20 years, since 2004.

2. Data Description

My data comes from the “**Billboard Year-End Hot 100 Singles USA.**” Most of the data was synthesized by a user on Kaggle, but it only went to 2021, so I manually added data from the

Billboard Year-End charts for the most recent years, which I considered important so we could truly see the impact period of before and *after*. Using Python and the Spotify API, I collected the durations of each song.

The data cleaning included extracting the song name, artist, chart year, and duration (in seconds). I decided to include duplicate songs across years because I am focused on what *types* of songs and durations people are listening to in any given year. The final sample was just the top 100 Year songs from the entire year, because I determined it would be more complicated to collect the data for any song that had charted on the Hot 100 during the entire year. Therefore, we are missing some popular songs, but we get a good idea of what the popular conscience in the U.S. was during each year. Out of the 2,100 songs, the Spotify API did not find 122 song durations. Therefore, the final sample has 1,978 songs and their durations since 2004.

3. Descriptive Statistics

Figure 3.1: Stata Summary of Data

Stata generated a table showing a detailed summary of all of the data, including information such as the mean (222.7), the standard deviation (43.3).

Figure 3.2: Histogram of Song Duration

Histogram showing a relatively normal distribution of all the song durations around the mean of 222.7.

Figure 3.3: Line Graph of Average Song Duration Per Year

Each data point represents the mean song duration of the Billboard End of Year Top 100 songs per year. The data shows a downward trend overall. From 2004 to 2017, the means fell from about 250

seconds to 220 seconds. However, around 2017, there was a stark decline before settling, staying relatively flat at around 200 seconds.

4. Analysis

Hypothesis Test:

For my hypothesis, I am going to use a two-sample t-test and compare the mean of the duration between two groups: songs before 2017, and songs after.

H_0 : There is no significant difference in song duration pre- and post-TikTok era (2017).

H_1 : Song durations have significantly decreased in length in the TikTok era (2017).

Or

$$H_0: \mu_0 (\text{Before}) - \mu_1 (\text{After}) = 0$$

$$H_1: \mu_0 (\text{Before}) - \mu_1 (\text{After}) \neq 0$$

Results: (See Figure 4.1: Stata Table of Two-Sample t-test)

The mean of songs before 2019 was 232.3 seconds based on 1,430 observations, 1.05 standard error, and a standard deviation of 39.9. The mean of songs after 2018 was 197.5 seconds with a 1.8 standard error, and a standard deviation of 41.7. This means that the difference between the two means is 34 seconds, which is a 17% reduction in overall song duration.

The t-test statistic is 17.17 with 1976 degrees of freedom. The p-value for this test was 0.0000, indicating that we should, with great certainty, reject the null hypothesis, meaning that songs are indeed shorter after the introduction of TikTok in 2018.

Linear Regression:

To observe a further relationship between year and song duration, I decided to run a linear regression:

$$duration = \beta_0 + \beta_1 * year + \varepsilon$$

(See Figure 4.2: Stata Regression Table)

The Regression Model:

$$duration = 5705.69 - 2.72 * year$$

The coefficient on year is -2.72, meaning that on average, song duration decreased by 2.72 seconds per year between 2004 and 2024. The p-value is 0.000, meaning that this result is highly statistically significant.

The R-squared is 0.1445, which means that about 14.45% of the variation in song duration can be explained by the year alone.

(See Figure 4.3: Scatter Graph with Regression Line)

5. Conclusion

Based on both the two-sample t-test and the linear regression analysis, there is strong statistical evidence that the average duration of the top songs in the United States has significantly decreased since the introduction of TikTok in 2018. The t-test confirmed a strong decline in song length with a p-value of 0.000, indicating significant evidence, while the regression analysis demonstrated a consistent downward trend over the past twenty years, with an annual decrease of approximately 2.7 seconds per year. These findings support my initial hypothesis that the rise of short-form content pioneered by TikTok has influenced not only how content is consumed but also how music is produced.

6. Work Cited

1. Kaggle Dataset, Billboard Year-End Hot 100 Singles USA 1946-2021 By User

philsong

<https://www.kaggle.com/datasets/liquidgenius1/billboard-yearend-hot-100-singles-usa>

2. Namu Wiki, Billboard Year-End Hot 100 Singles USA

<https://en.namu.wiki/w/%EB%B9%8C%EB%B3%B4%EB%93%9C%20%EC%97%B0%EB%A7%90%20%EC%B0%A8%ED%8A%B8#s-4.8>

3. Spotify API,

<https://developer.spotify.com/documentation/web-api>

4. Chat GPT, OpenAI

<https://chatgpt.com/>

*Used to help create Python Code for extracting durations from the Spotify API

7. Appendix

Figure 3.1: Stata Summary of Data

Duration_seconds				
Percentiles		Smallest		
1%	128.013	80.927		
5%	161.384	109.75		
10%	176.546	114.233	Obs	1,978
25%	198.16	114.893	Sum of wgt.	1,978
50%	220.496	Largest	Mean	222.6746
75%	242.373		Std. dev.	43.28538
90%	269.666	503.771	Variance	1873.624
95%	288.76	613.026	Skewness	1.832363
99%	338.853	688.453	Kurtosis	16.51622

Figure 3.2: Histogram of Song Duration

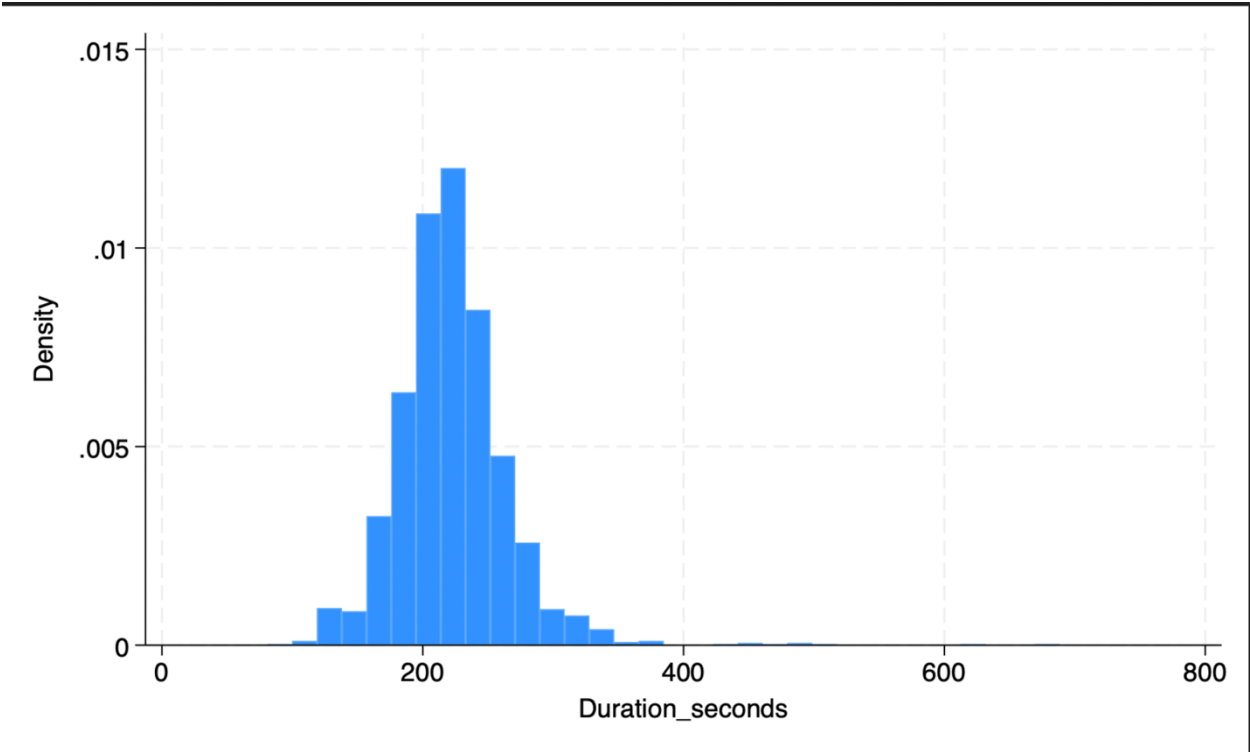


Figure 3.3: Line Graph of Average Song Duration Per Year

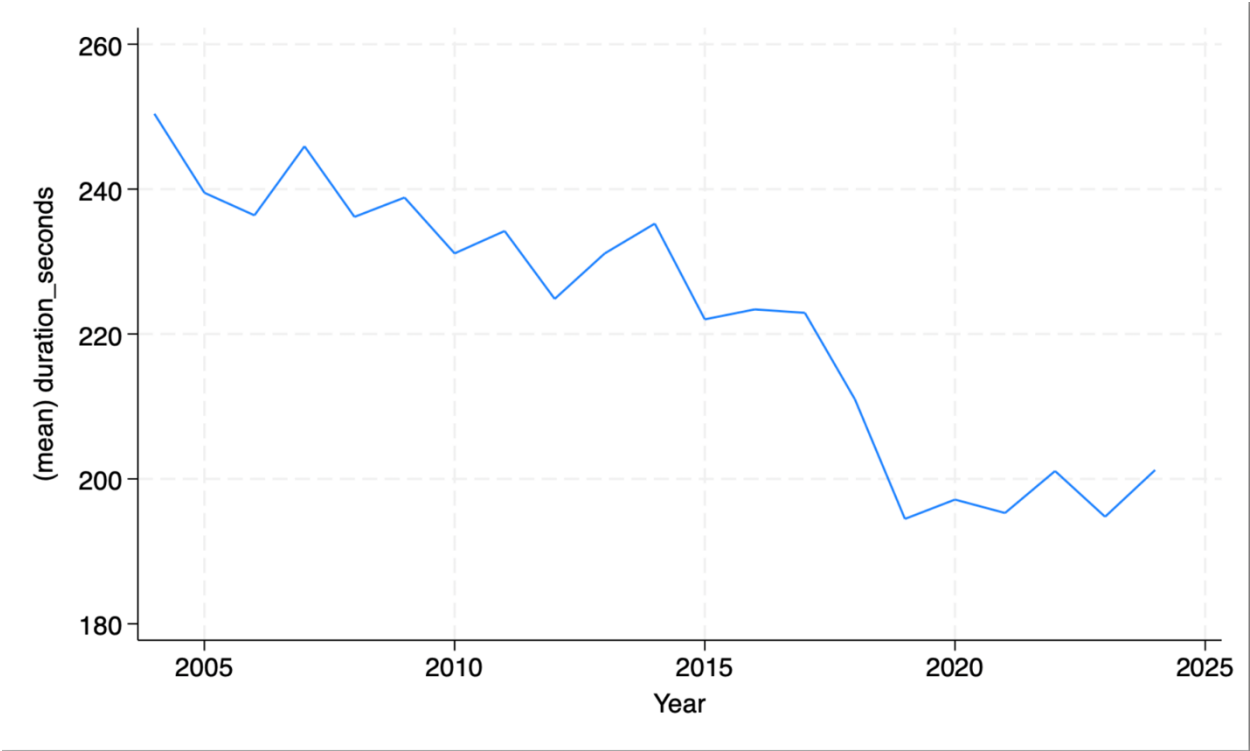


Figure 4.1: Stata Table of Two-Sample t-test

Two-sample t test with equal variances						
Group	Obs	Mean	Std. err.	Std. dev.	[95% conf. interval]	
0	1,430	232.3306	1.053954	39.85563	230.2631	234.3981
1	548	197.4772	1.783019	41.7394	193.9748	200.9797
Combined	1,978	222.6746	.9732583	43.28538	220.7658	224.5833
diff		34.85335	2.029011		30.87413	38.83258
diff = mean(0) - mean(1)				t =	17.1775	
H0: diff = 0				Degrees of freedom =	1976	
Ha: diff < 0		Ha: diff != 0		Ha: diff > 0		
Pr(T < t) = 1.0000		Pr(T > t) = 0.0000		Pr(T > t) = 0.0000		

Figure 4.2: Stata Regression Table

regress duration year

Source	SS	df	MS	Number of obs	=	1,978
Model	535228.294	1	535228.294	F(1, 1976)	=	333.74
Residual	3168927.06	1,976	1603.70802	Prob > F	=	0.0000
Total	3704155.35	1,977	1873.62436	R-squared	=	0.1445
				Adj R-squared	=	0.1441
				Root MSE	=	40.046

duration_s~s	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
year	-2.722583	.1490302	-18.27	0.000	-3.014856	-2.43031
_cons	5705.687	300.1334	19.01	0.000	5117.076	6294.298

Figure 4.3: Scatter Graph with Regression Line

